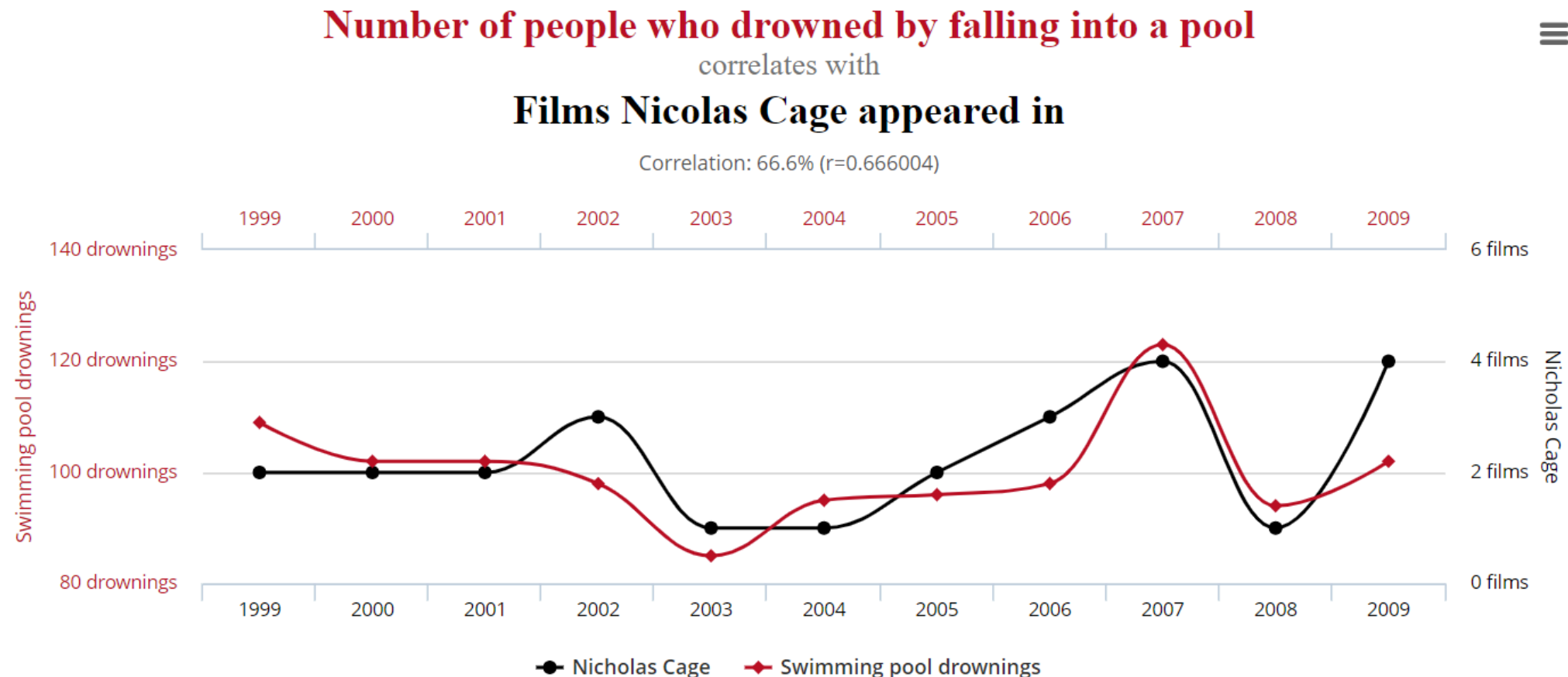


DS 3100

Fundamentals of Data Science

Advanced Correlation Analysis



Learning Goals

- **Use correlation for exploratory and inferential analysis**
- **Know the limitations of correlation analyses**
- **Learn different types of correlations and when to use them**
 - Pearson's correlation
 - Point-biserial correlation
 - Tetrachoric correlation

Explore bivariate associations

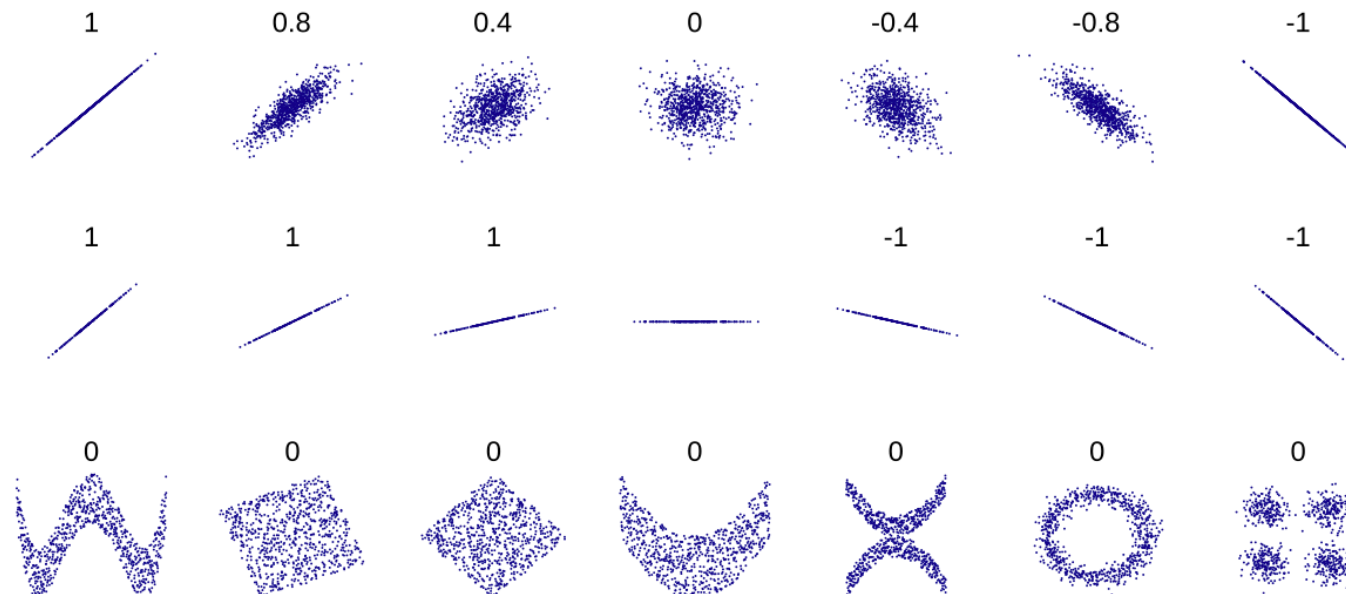
Infer relations to motivate analyses

Conceptual Basis



Definition: a correlation is a **standardized** measure of the **strength** (value) and **direction** (sign) of a **linear** relationship between **two** variables

<https://www.guessthecorrelation.com>



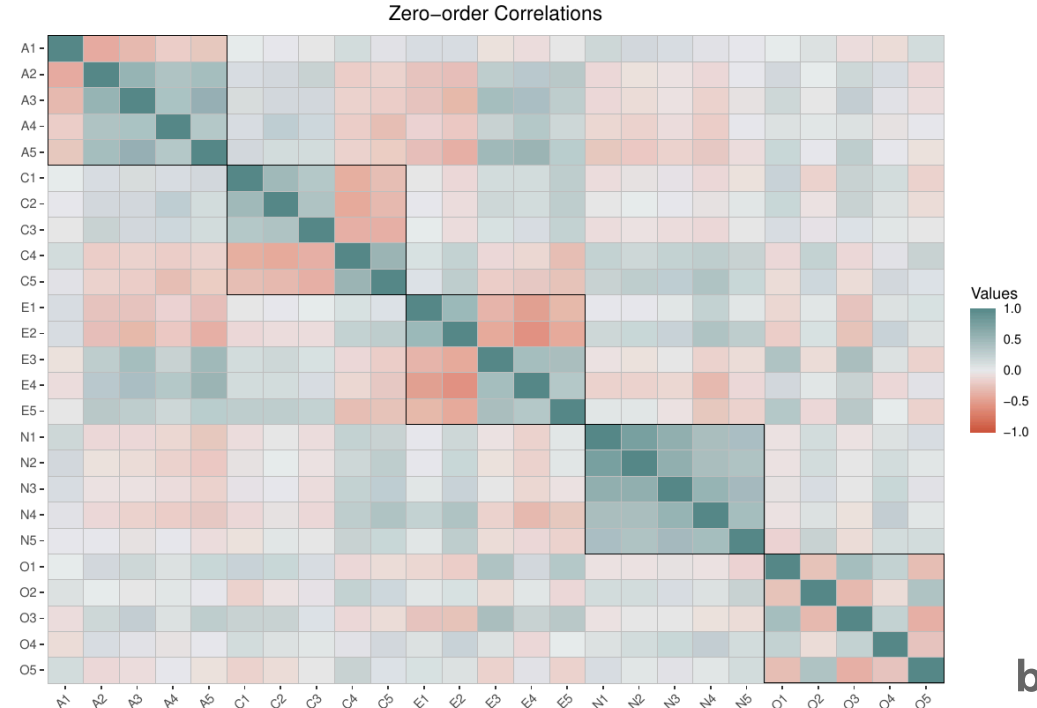
Correlation Matrix

table showing bivariate relationships
across two or more variables

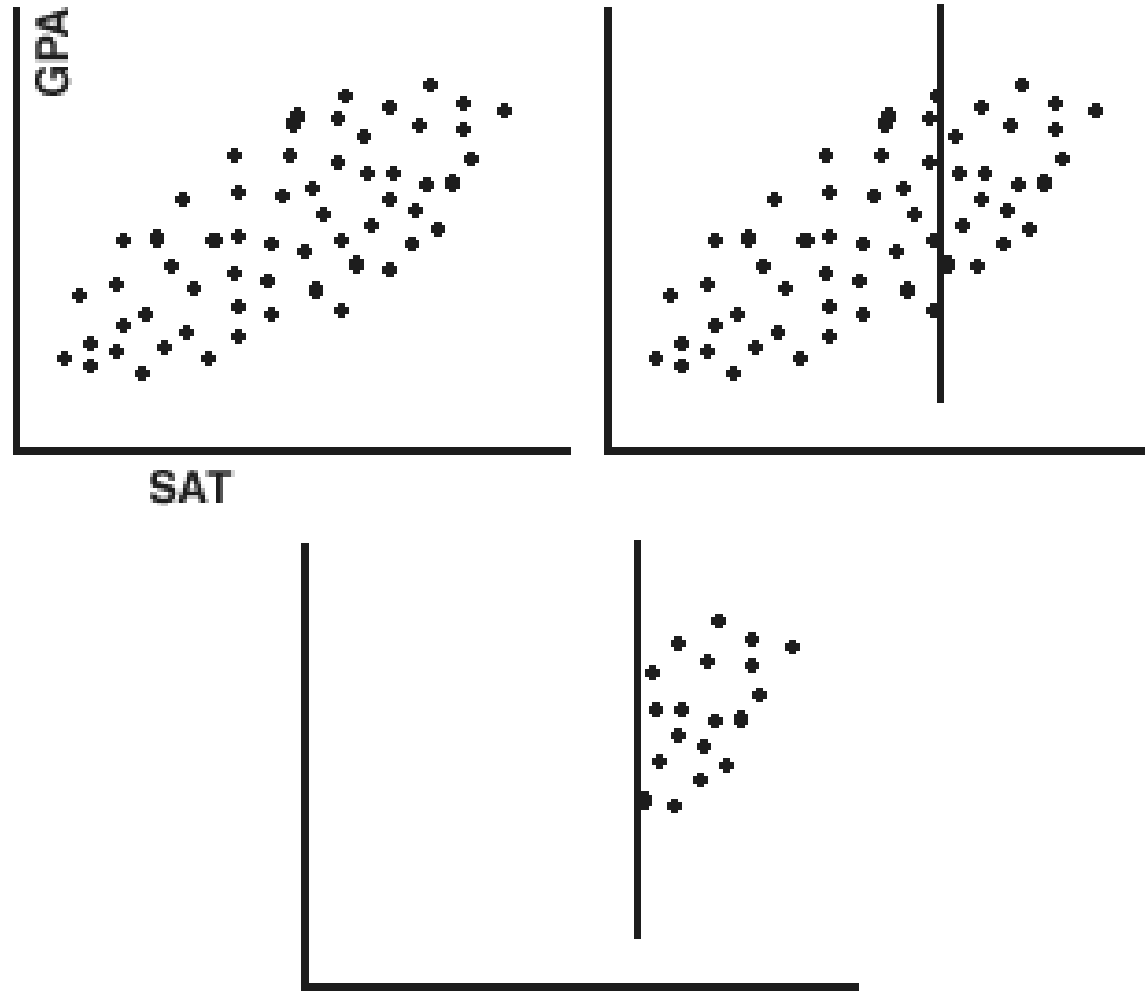
Properties

- symmetric: upper and lower diagonals are identical
 - Diagonal elements = 1
- (CORRELATIONS OF VARIABLES WITH THEMSELVES)

	A1	A2	A3	A4	A5	C1	C2	C3	C4	C5	E1	E2	E3	E4	E5	N1	N2	N3	N4	N5	O1	O2	O3	O4	O5
A1	1.00	-0.41	-0.32	-0.18	-0.23	0.00	0.01	-0.02	0.15	0.06	0.12	0.11	-0.07	-0.08	-0.02	0.18	0.16	0.11	0.06	0.01	0.00	0.08	-0.09	-0.10	0.14
A2	-0.41	1.00	0.56	0.39	0.45	0.12	0.16	0.23	-0.19	-0.16	-0.24	-0.27	0.29	0.32	0.32	-0.11	-0.07	-0.05	-0.12	0.02	0.16	0.00	0.19	0.11	-0.11
A3	-0.32	0.56	1.00	0.41	0.57	0.12	0.16	0.16	-0.16	-0.19	-0.25	-0.33	0.45	0.42	0.29	-0.11	-0.10	-0.06	-0.15	-0.05	0.19	-0.03	0.27	0.06	-0.08
A4	-0.18	0.39	0.41	1.00	0.36	0.11	0.27	0.17	-0.20	-0.28	-0.14	-0.21	0.22	0.35	0.19	-0.12	-0.16	-0.07	-0.19	-0.01	0.08	0.03	0.08	-0.04	0.02
A5	-0.23	0.45	0.57	0.36	1.00	0.16	0.14	0.15	-0.17	-0.20	-0.28	-0.38	0.48	0.53	0.30	-0.23	-0.21	-0.16	-0.22	-0.09	0.20	-0.01	0.27	0.02	-0.07
C1	0.00	0.12	0.12	0.11	0.16	1.00	0.48	0.34	-0.40	-0.29	-0.03	-0.11	0.14	0.15	0.28	-0.08	-0.05	-0.04	-0.11	-0.07	0.21	-0.14	0.22	0.14	-0.16
C2	0.01	0.16	0.16	0.27	0.14	0.48	1.00	0.40	-0.43	-0.33	0.01	-0.07	0.18	0.14	0.29	-0.02	0.00	0.01	-0.04	0.05	0.20	-0.06	0.22	0.07	-0.09
C3	-0.02	0.23	0.16	0.17	0.15	0.34	0.40	1.00	-0.38	0.00	-0.09	0.10	0.11	0.25	-0.07	-0.06	-0.08	-0.12	0.02	0.11	-0.04	0.07	0.03	-0.02	
C4	0.15	-0.19	-0.16	-0.20	-0.17	-0.40	-0.43	-0.38	1.00	0.53	0.10	0.23	-0.10	-0.13	-0.28	0.25	0.18	0.24	0.29	0.23	-0.11	0.25	-0.10	0.06	0.23
C5	0.06	-0.16	-0.19	-0.28	-0.20	-0.29	-0.33	-0.38	0.53	1.00	0.07	0.29	-0.18	-0.22	-0.27	0.23	0.27	0.27	0.38	0.20	-0.10	0.17	-0.10	0.16	0.07
E1	0.12	-0.24	-0.25	-0.14	-0.28	-0.03	0.01	0.00	0.10	0.07	1.00	0.52	-0.35	-0.48	-0.33	0.02	0.01	0.05	0.25	0.05	-0.12	0.05	-0.24	0.09	0.10
E2	0.11	-0.27	-0.33	-0.21	-0.38	-0.11	-0.07	-0.09	0.23	0.29	0.52	1.00	-0.42	-0.58	-0.42	0.18	0.21	0.21	0.38	0.28	-0.19	0.10	-0.26	0.21	0.09
E3	-0.07	0.29	0.45	0.22	0.48	0.14	0.18	0.10	-0.10	-0.18	-0.35	-0.42	1.00	0.46	0.43	-0.06	-0.07	-0.02	-0.16	-0.09	0.38	-0.09	0.44	0.08	-0.15
E4	-0.08	0.32	0.42	0.35	0.53	0.15	0.14	0.11	-0.13	-0.22	-0.48	-0.58	0.46	1.00	0.35	-0.15	-0.15	-0.12	-0.32	-0.11	0.17	0.05	0.23	-0.12	0.06
E5	-0.02	0.32	0.29	0.19	0.30	0.28	0.29	0.25	-0.28	-0.27	-0.33	-0.42	0.43	0.35	1.00	0.05	0.05	-0.06	-0.23	-0.15	0.35	-0.11	0.33	0.00	-0.15
N1	0.18	-0.11	-0.11	-0.12	-0.23	-0.08	-0.02	-0.07	0.25	0.23	0.02	0.18	-0.06	-0.15	0.05	1.00	0.76	1.00	0.60	0.43	0.39	-0.05	0.14	-0.03	0.16
N2	0.16	-0.07	-0.10	-0.16	-0.21	-0.05	0.00	-0.06	0.18	0.27	0.01	0.21	-0.07	-0.15	0.05	0.76	1.00	0.60	0.43	0.39	-0.05	0.14	-0.03	0.16	0.04
N3	0.11	-0.05	-0.06	-0.07	-0.16	-0.04	0.01	-0.08	0.24	0.27	0.05	0.21	-0.02	-0.12	-0.06	0.61	0.60	1.00	0.56	0.47	-0.04	0.12	-0.03	0.21	0.06
N4	0.06	-0.12	-0.15	-0.19	-0.22	-0.11	-0.04	-0.12	0.29	0.38	0.25	0.38	-0.16	-0.32	-0.23	0.44	0.43	0.56	1.00	0.44	-0.05	0.09	-0.07	0.26	0.04
N5	0.01	0.02	-0.05	-0.01	-0.09	-0.07	0.05	-0.02	0.23	0.20	0.05	0.28	-0.09	-0.11	-0.15	0.42	0.39	0.47	0.44	1.00	-0.14	0.23	-0.10	0.14	0.16
O1	0.00	0.16	0.19	0.08	0.20	0.21	0.20	0.11	-0.11	-0.10	-0.12	-0.19	0.38	0.17	0.35	-0.06	-0.05	-0.04	-0.05	-0.14	1.00	-0.26	0.45	0.25	-0.30
O2	0.08	0.00	-0.03	0.03	-0.01	-0.14	-0.06	-0.04	0.25	0.17	0.05	0.10	-0.09	0.05	-0.11	0.14	0.14	0.12	0.09	0.23	-0.26	1.00	-0.32	-0.10	0.37
O3	-0.09	0.19	0.27	0.08	0.27	0.22	0.22	0.07	-0.10	-0.10	-0.24	-0.26	0.44	0.23	0.33	-0.06	-0.03	-0.03	-0.07	-0.10	0.45	-0.32	1.00	0.25	-0.37
O4	-0.10	0.11	0.06	-0.04	0.02	0.14	0.07	0.03	0.06	0.16	0.09	0.21	0.08	-0.12	0.00	0.09	0.16	0.21	0.26	0.14	0.25	-0.10	0.25	1.00	-0.24
O5	0.14	-0.11	-0.08	0.02	-0.07	-0.16	-0.09	-0.02	0.23	0.07	0.10	0.09	-0.15	0.06	-0.15	0.12	0.04	0.06	0.04	0.16	-0.30	0.37	-0.37	-0.24	1.00

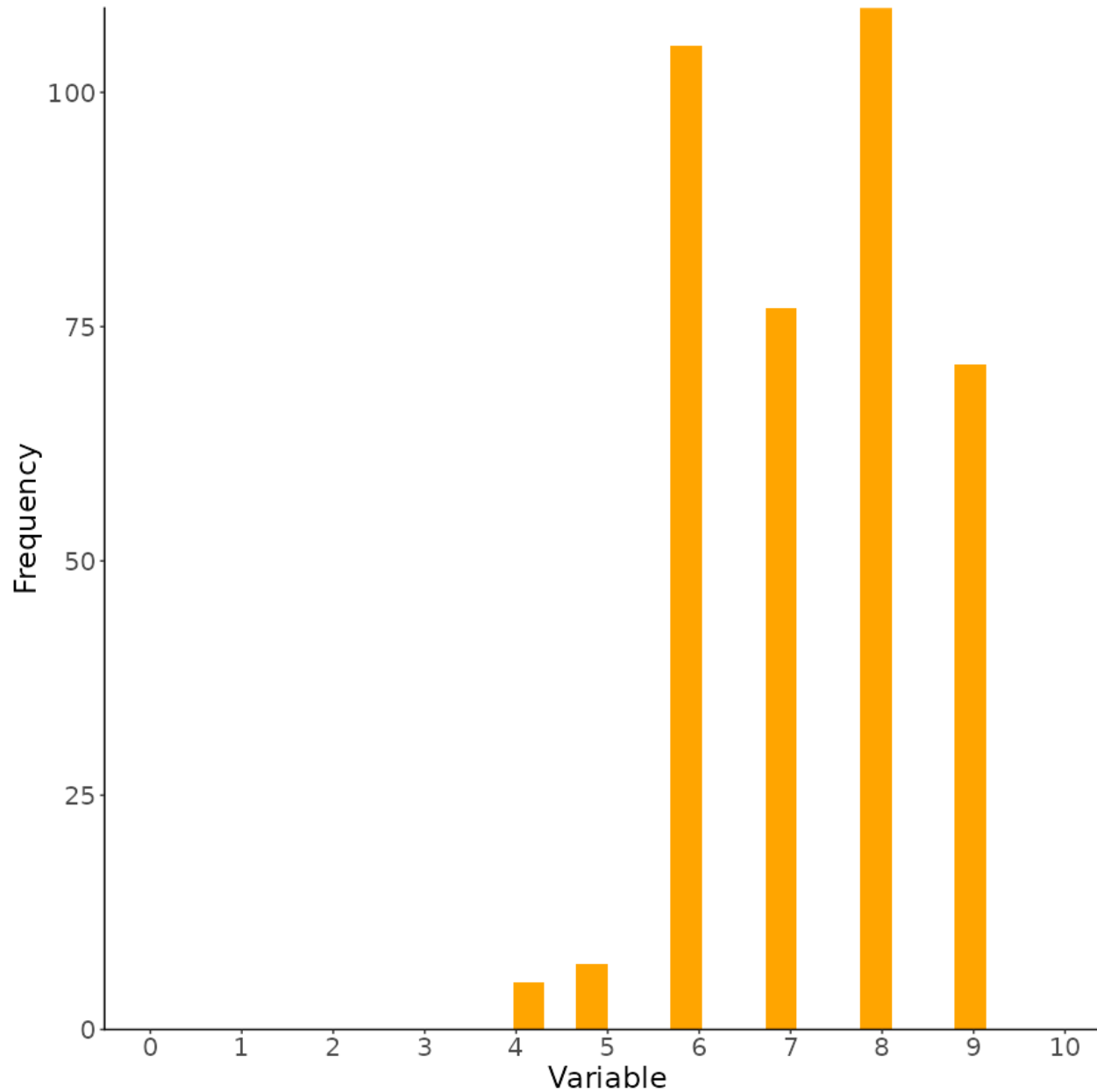


Things to Consider



Restriction of Range

Things to Consider

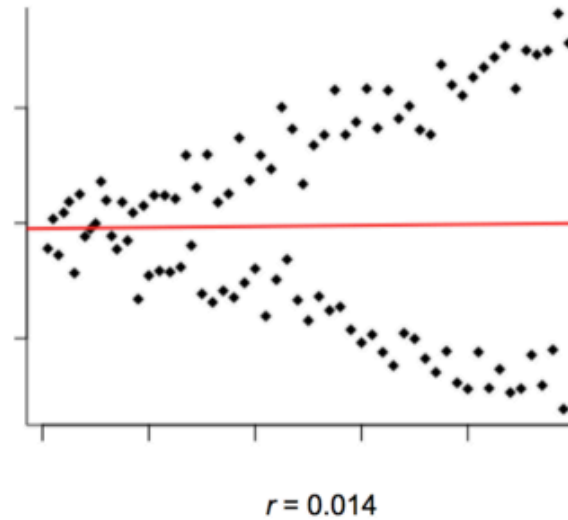
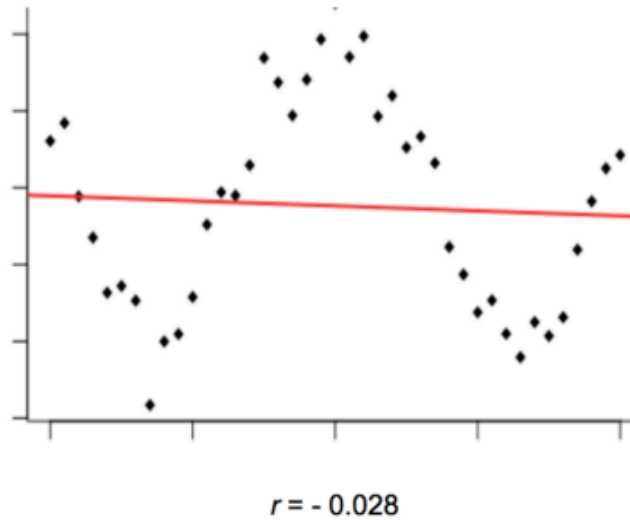


Number of Actual Values vs. Theoretical

Just because there are 10 values *theoretically* does not mean there are 10 values

This variable should be considered **ordinal** *not* continuous

Things to Consider



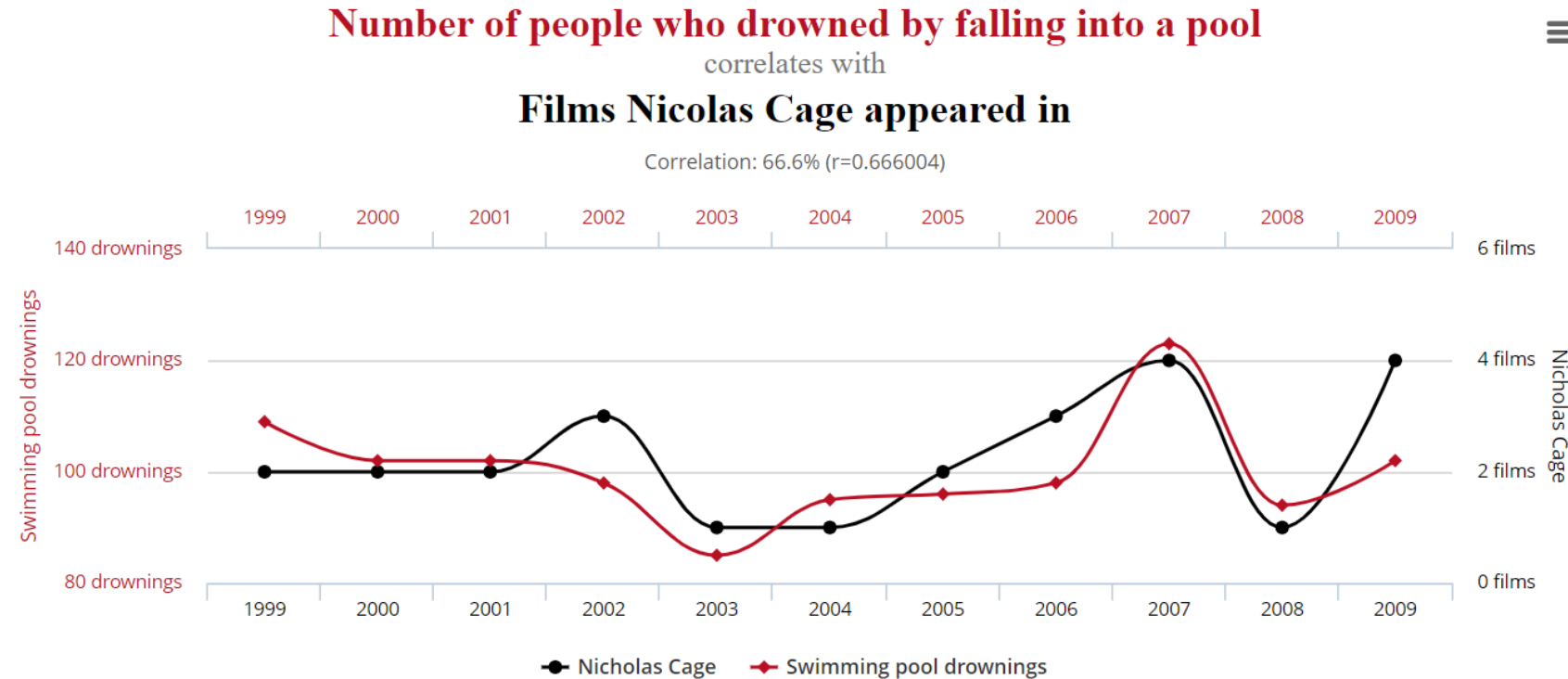
Nonlinear Relations

Cannot accurately capture relationships that are *not* linear

Spurious Correlations

Relationships that are not actually related but simply due to chance patterns

Things to Consider





Measurement Scales

There is more than *one* correlation

Scales of Measurement

Ways to classify data into different uses (especially for visualization and analysis)

Nominal

identity, name, and/or categorize
something or someone
does not indicate order

e.g., animals, country of
residence, Olympic sport

Ordinal

convey order or rank

no equal spacing between
categories

e.g., highest awarded degree,
sports rankings, disagree-agree

Interval

equal spacing in units (no true
zero)

differences are comparable

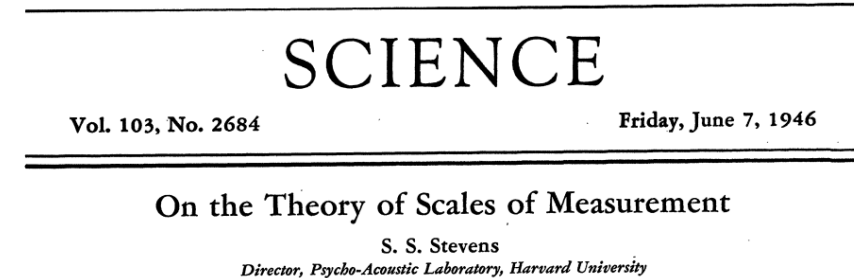
e.g., Celsius/Fahrenheit, calendar
years, pH

Ratio

equal spacing in units (and
true zero)

comparable

e.g., Kelvin, weight, speed

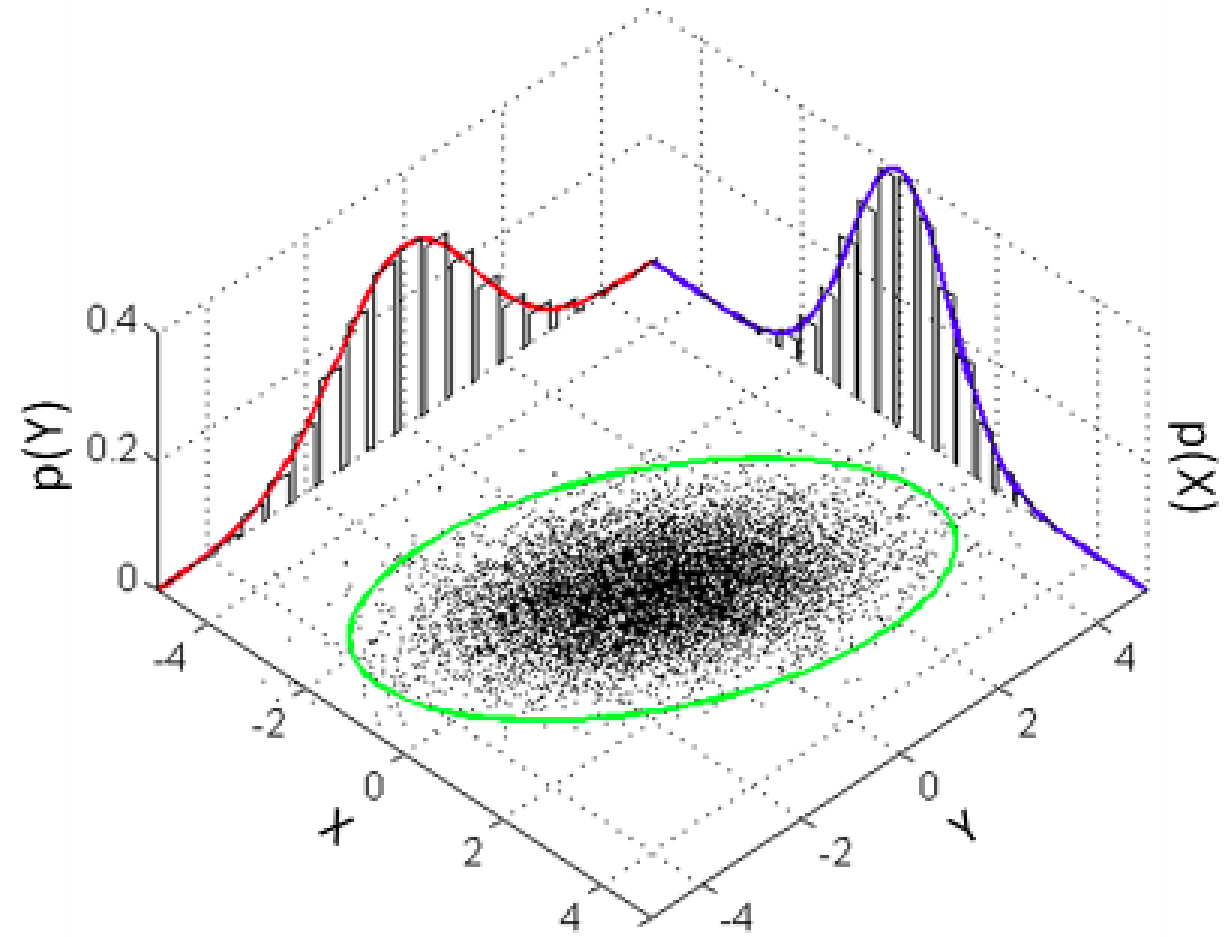


Types of Correlation



Pearson's Correlation

between two interval and/or ratio scale variables



Pearson's Correlation

z-score form

$$Z = \frac{x - \mu}{\sigma}$$

$$r = 1 - \frac{\sum (z_x - z_y)^2}{2(n - 1)}$$

average squared distance between z-scores

$$r = \frac{\sum z_x z_y}{n - 1}$$

average cross-product of z-scores

Part 1: Pearson's Correlation

- Data were collected at CERN and obtained via the [open source data repository](#)
- Data measure energy and momentum of muons, a subatomic particle and basic building block of the universe

$$E^2 = (mc^2)^2 + (pc)^2$$

energy mass speed of light momentum

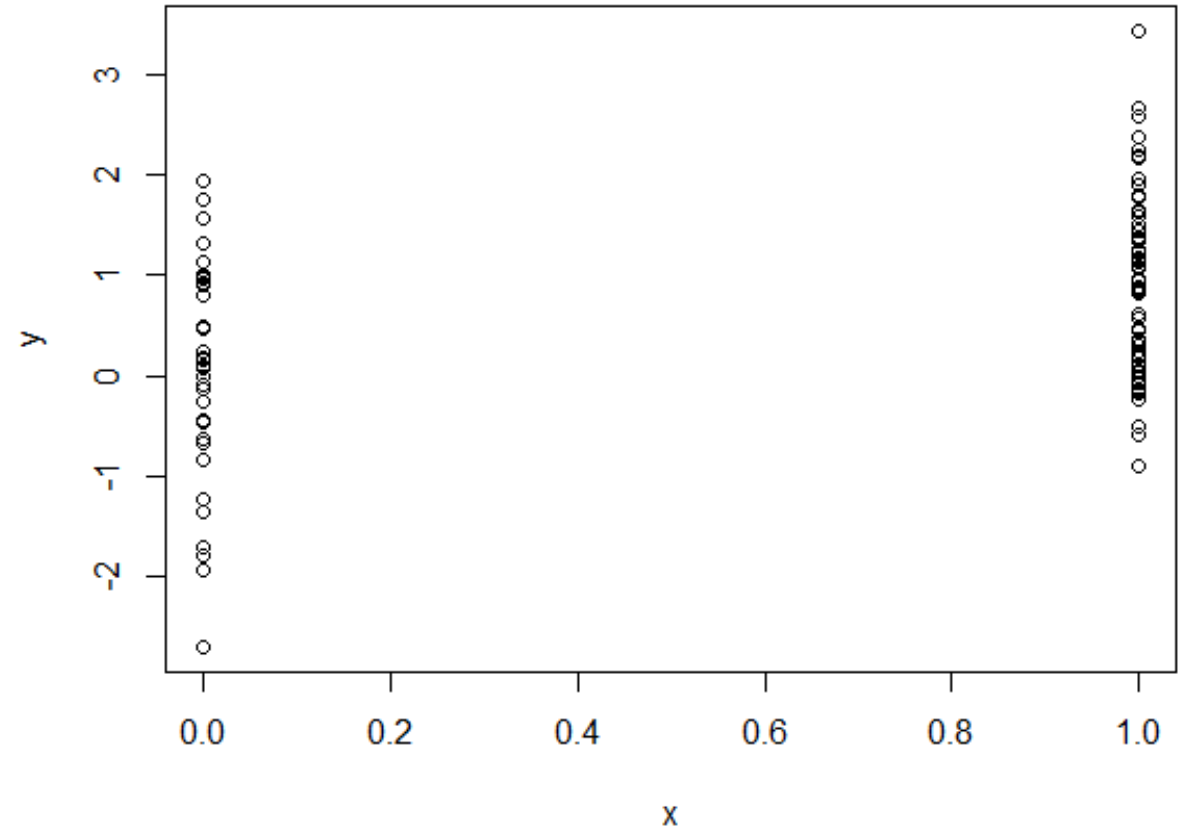
For particles with very little mass, is there a relationship between energy and momentum?



Point-biserial Correlation

between one interval or ratio variable and one
nominal or ordinal variable

must have two categories **only** in nominal/ordinal
variable



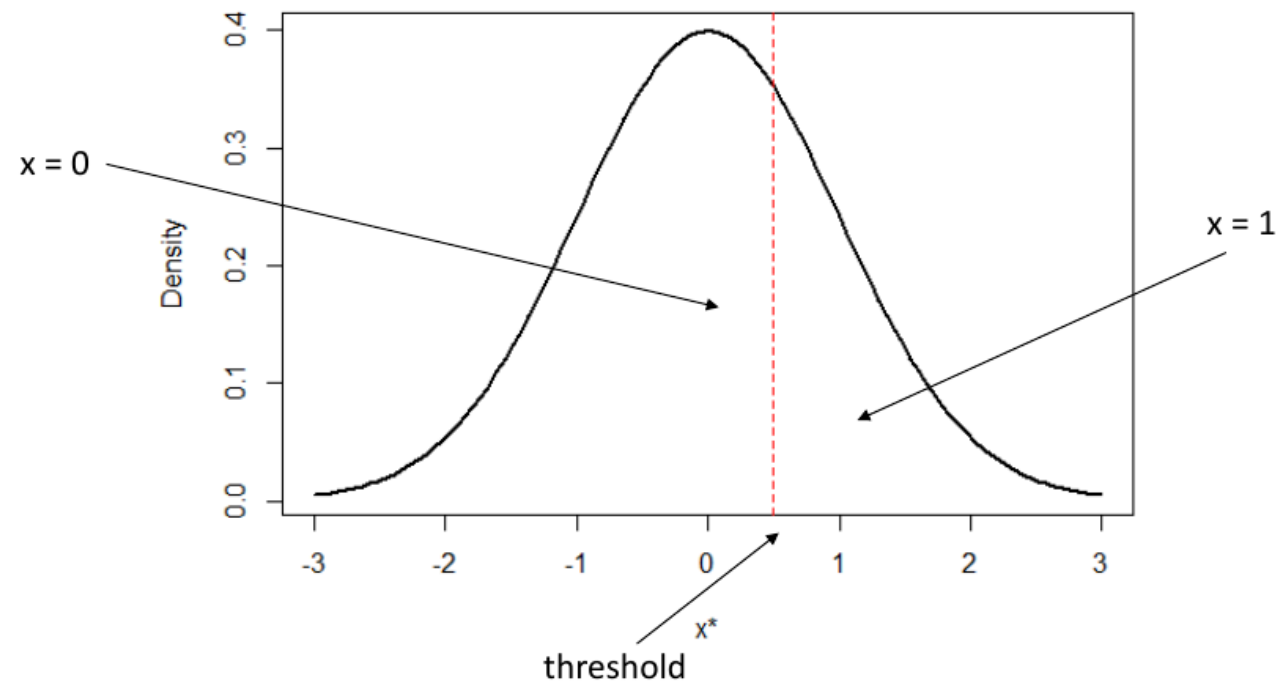
Tetrachoric Correlation

between two nominal/ordinal variables
must have two categories **only** in each

	X = 0	X = 1
Y = 0	38	10
Y = 1	18	34

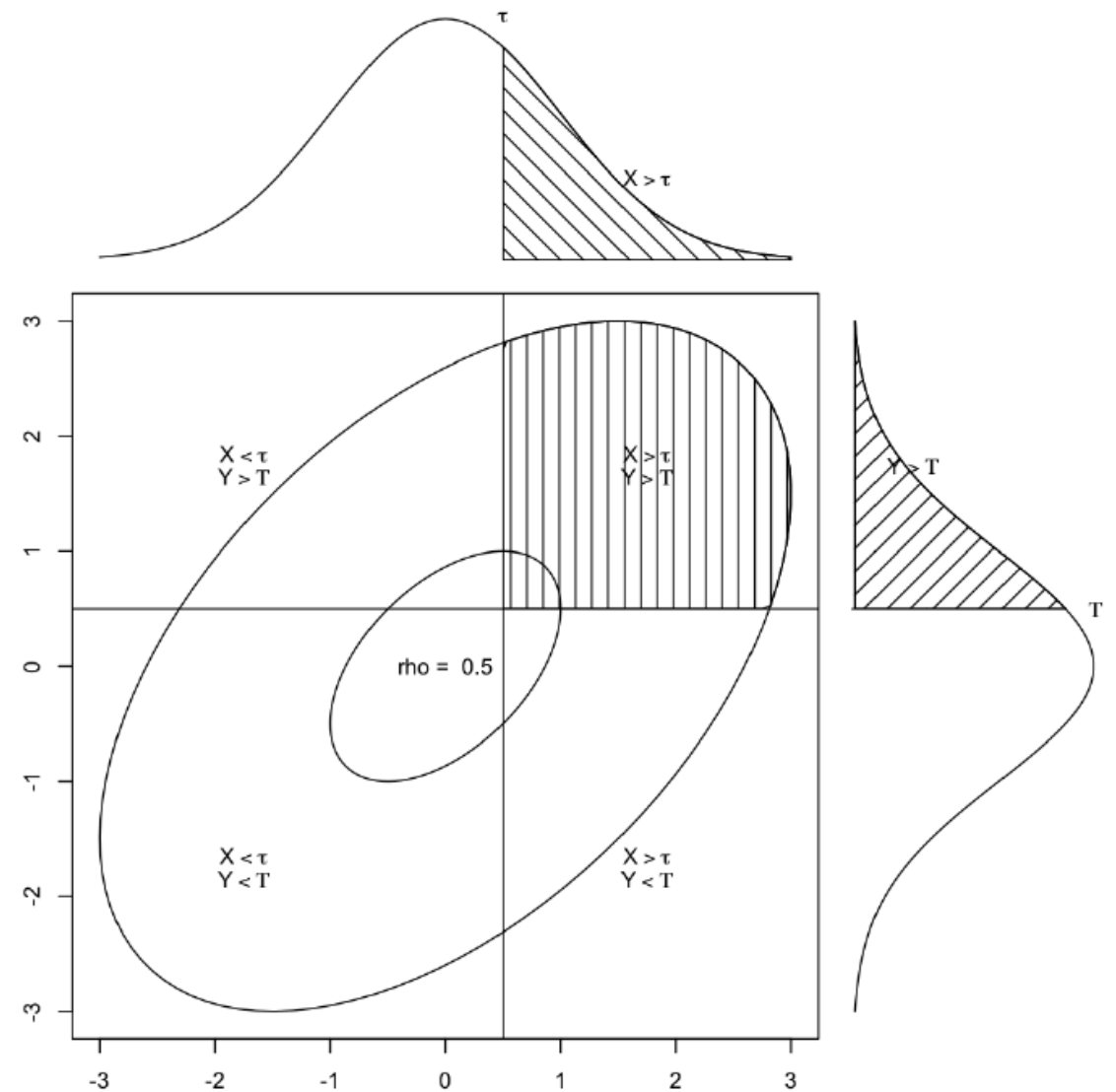
Tetrachoric Correlation

continuous conceptualization of the data



Tetrachoric Correlation

continuous conceptualization of the data



Part 2: Tetrachoric Correlation

- Web scraped from [Rate My Professor](#) (2020)

QUALITY

4.0

DIFFICULTY

3.5

CHM110 - Introduction to Chemistry

Textbook: **No**

Great teacher but we had to go online half-way through the semester because his ex-husband shot him in the leg. Obviously not his fault, but it was pretty disruptive and his in-person lectures were better than his online ones. Would take again now that his ex is out of the picture.

He seems happy, and teaches well, but there's just something off about him that I can't put my finger on. He smiles, but there's no warmth there. Just a terrible emptiness like you'd find in the rusting hull of a ship forgotten at the bottom of a dead sea. Something happened to him to make him this way. I do not know what it was. No one does.

 Report this rating

FST1000

Dec 23rd, 2023

QUALITY

5.0

DIFFICULTY

1.0

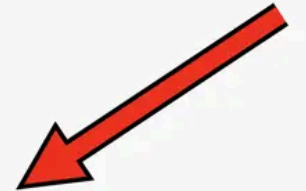
For Credit: **Yes**

Attendance: **Mandatory**

Would Take Again: **Yes**

Grade: **A**

Textbook: **N/A**



i would give her my kidney

CLEAR GRADING CRITERIA

INSPIRATIONAL

CARING

MADE WITH

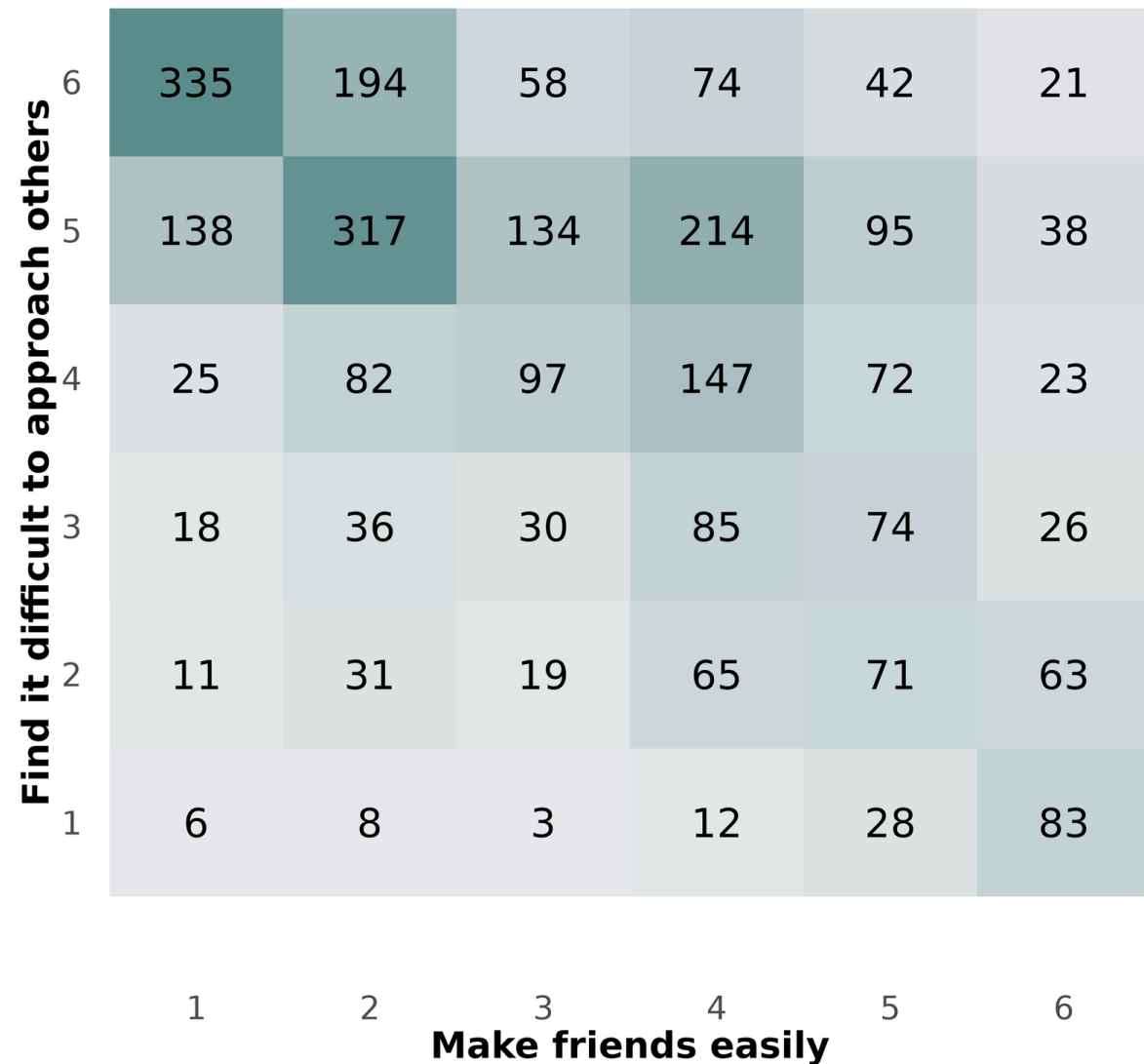
beautiful.ai

Your Turn

- **All variables are self-evident and coded as 0 (does not have tag) and 1 (has tag)**
- Come up with a research question that looks at the association between two variables
- Write the formal hypothesis
- Make a prediction (using effect size)
- Compute the tetrachoric correlation
- Interpret and reflect on your result

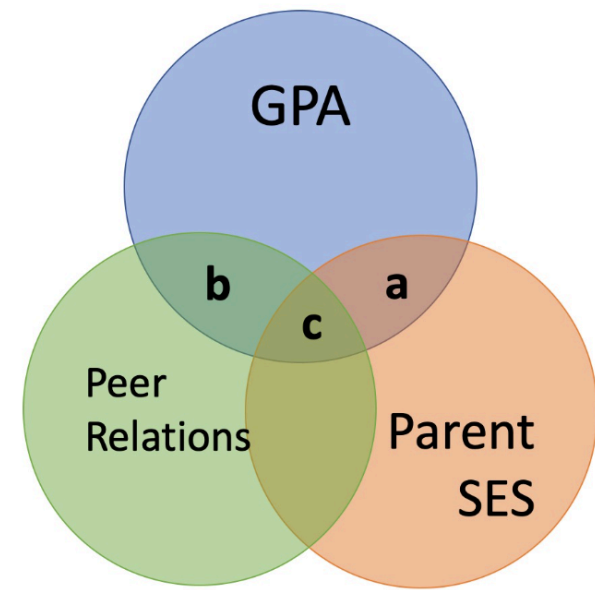
Polychoric Correlation

between two nominal/ordinal variables
must have more than two categories in each

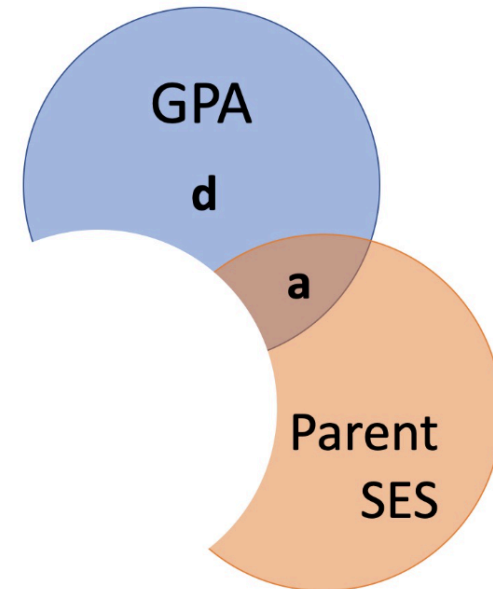


Partial Correlation

between two interval and/or ratio scale variables
given some third (or more) variable



Pearson's Correlation



Partial Correlation

Partial Correlation

form

Z = all variables being controlled for

$$\rho_{XY \cdot Z} = \frac{\rho_{XY} - \rho_{XZ}\rho_{ZY}}{\sqrt{1 - \rho_{XZ}^2} \sqrt{1 - \rho_{ZY}^2}}$$

Part 3: Partial Correlation

- Data were collected by the World Health Organization and obtained via [Kaggle](#)
 - 183 countries (rows)
 - 22 health-related variables
 - Only data from 2014



Is there a relationship between population levels of substance use on life expectancy *given* GDP (expenditure)?

Variable Documentation

- Country
- Year (all 2014)
- Status (developed versus developing)
- **Life.expectancy (in years)**
- Adult.Mortality (probability of dying between 15–60 per 1000)
- **Alcohol (per capita consumption in liters of alcohol)**
- **percentage.expenditure (percentage GDP spent on health)**
- and others....